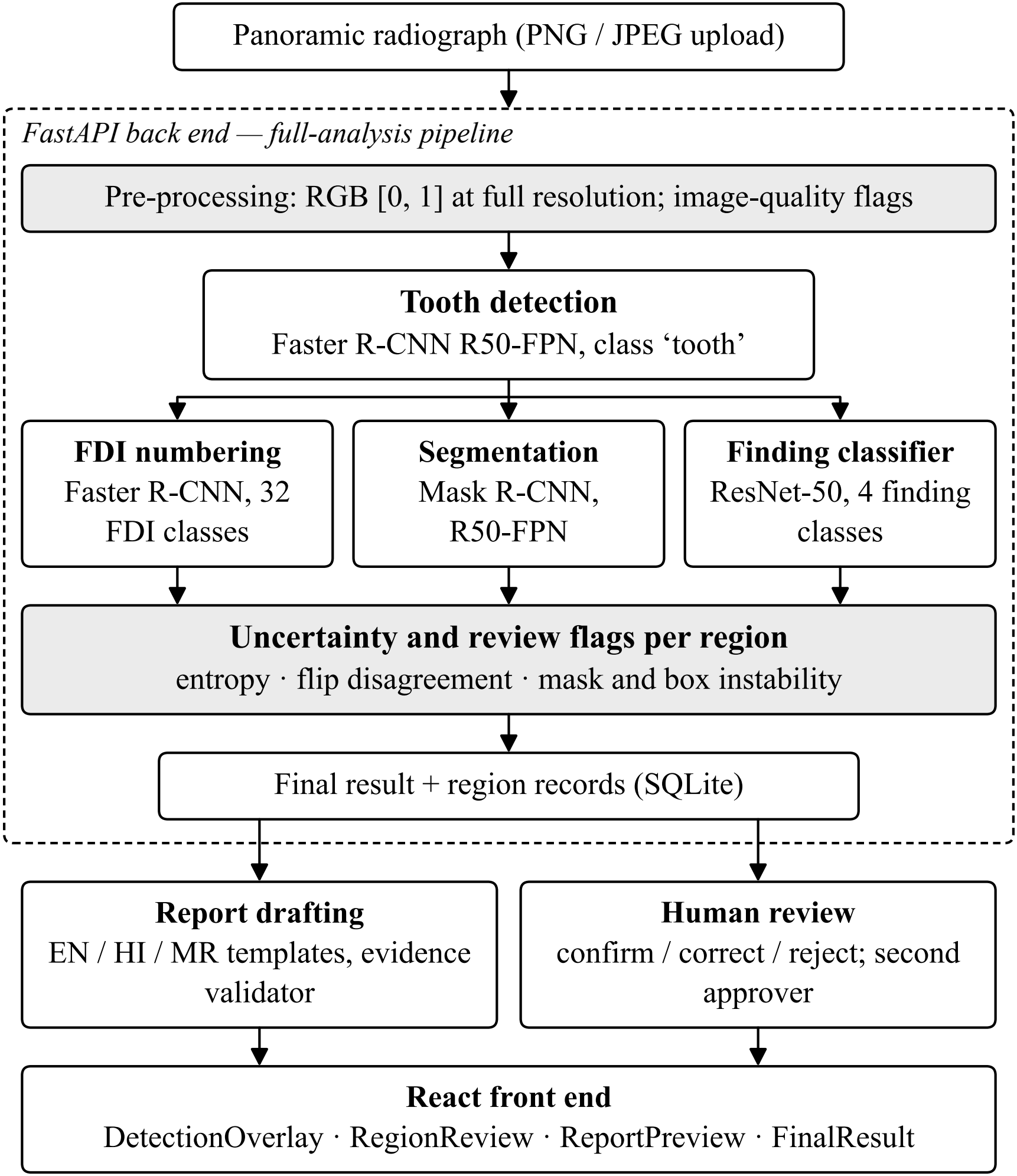

Panoramic radiograph (PNG / JPEG upload)
FastAPI back end — full-analysis pipeline
Pre-processing: RGB [0, 1] at full resolution; image-quality flags
Tooth detection
Faster R-CNN R50-FPN, class 'tooth'
FDI numbering
Faster R-CNN, 32 FDI classes
Segmentation
Mask R-CNN, R50-FPN
Finding classifier
ResNet-50, 4 finding classes
Uncertainty and review flags per region
entropy · flip disagreement · mask and box instability
Final result + region records (SQLite)
Report drafting
EN / HI / MR templates, evidence validator
Human review
confirm / correct / reject; second approver
React front end
DetectionOverlay · RegionReview · ReportPreview · FinalResult